multi_agent_financial_analysis

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1 Multi-Agent Financial Analysis System

AAI-520 Final Team Project — Brandon Wirgau, Christina, and Isaac

This notebook is the canonical, executable report for an agentic investment-research system. It accepts a stock ticker and user research question, uses LangGraph to route a bounded run to market, financial, news, or comprehensive specialists, evaluates and revises the complete report, and stores reusable lessons for later runs.

Repository: AAI-520 Final Project

> Research support only. The output is not personalized investment advice and does not guarantee future performance.

1.1 1. Project objectives and implemented capabilities

The project demonstrates four autonomous agent functions:

1. **Plan** research objectives, questions, and tool selection using prior-run memory as context.
2. **Use tools** to retrieve Yahoo Finance market/fundamental data, SEC EDGAR facts and filings, and multi-source company news.
3. **Reflect and evaluate** data quality, report grounding, and specialist-analysis completeness.
4. **Learn across runs** by persisting bounded, timestamped research lessons.

It also implements all three required workflow patterns:

- **Prompt chaining:** ingest → preprocess → classify → extract → summarize.
- **Routing:** the user request selects market-only, financial-only, news-only, or comprehensive research; news articles also route to earnings, market, and general-news specialists.
- **Evaluator–optimizer:** article analysis uses a four-part rubric, while the complete report is scored for completeness, grounding, reasoning, consistency, sources, and clarity and revised through a bounded LangGraph loop.

The refactored modules preserve the original notebook's planning, tool execution, validation, reflection, reporting, memory, SEC EDGAR retrieval, 10-year price graph, multi-source news analysis, specialist routing, and iterative optimization. The updated Google News RSS fallback and retry-aware HTTP behavior are included as well.

1.2 2. Implemented system architecture

The research workflow keeps provider adapters, provider-independent tools, agents, workflows, memory, and presentation concerns separate. This makes each stage inspectable and independently

testable.

```
flowchart LR
    U[User ticker and research question] --> P[Planner]
    L[Research memory] --> P
    P --> RQ{Request route}
    RQ -->|Market| MA[Market specialist]
    RQ -->|Financial| FA[Financial specialist]
    RQ -->|News| C[News prompt chain]
    RQ -->|Comprehensive| MA
    RQ -->|Comprehensive| FA
    RQ -->|Comprehensive| C
    Y[Yahoo Finance] --> MA
    Y --> FA
    S[SEC EDGAR] --> FA
    Y --> C
    N[NewsAPI] --> C
    G[Google News RSS fallback] --> C
    C --> R[Specialist router]
    R --> E[Earnings specialist]
    R --> M[Market specialist]
    R --> W[News specialist]
    E --> V[Evaluator / optimizer]
    M --> V
    W --> V
    MA --> Z[Synthesizer]
    FA --> Z
    V --> Z
    Z --> Q{Six-part report evaluator}
    Q -->|Below threshold| O[Report optimizer]
    O --> Q
    Q -->|Pass or revision limit| F[Final report and visuals]
    F --> L
```

1.3 3. Environment setup

Create and activate a virtual environment, install `requirements.txt`, start Ollama, and copy `.env.example` to `.env`. At minimum:

- `OLLAMA_MODEL=llama3.2` and a running `ollama serve` process;
- `SEC_USER_AGENT` containing an application name and monitored contact email;
- optional `NEWS_API_KEY` (Yahoo remains primary and Google RSS is a fallback);
- optional `DEFAULT_TICKER`, `RESEARCH_REQUEST`, `MAX_NEWS_ARTICLES`, `EVALUATION_SCORE_THRESHOLD`, `MAX_OPTIMIZER_ITERATIONS`, `REPORT_SCORE_THRESHOLD`, and `MAX_REPORT_REVISIONS`.

Credentials are read only from environment variables and are never displayed below.

[1]:
```
import getpass
import json
import os
import shutil
import subprocess
import sys
import tempfile
import time
import urllib.request
from pathlib import Path

try:
    import google.colab  # noqa: F401

    IN_COLAB = True
except ImportError:
    IN_COLAB = False

if IN_COLAB:
    repo_url = "https://github.com/bwirgau-usd/
 ↪AAI-520-final-project-multi-agent-financial-analysis.git"
    repo_dir = Path("/content/
 ↪AAI-520-final-project-multi-agent-financial-analysis")
    if not repo_dir.exists():
        subprocess.run(
            ["git", "clone", "--branch", "main", repo_url, str(repo_dir)],
            check=True,
        )
    os.chdir(repo_dir)
    subprocess.run(
        [sys.executable, "-m", "pip", "install", "-q", "-r", "requirements.
 ↪txt"],
        check=True,
    )

    if not os.getenv("NEWS_API_KEY"):
        news_key = getpass.getpass(
            "NEWS_API_KEY (optional; input hidden): "
        ).strip()
        if news_key:
            os.environ["NEWS_API_KEY"] = news_key
    if not os.getenv("SEC_USER_AGENT"):
        os.environ["SEC_USER_AGENT"] = input(
            "SEC user agent (name and contact email): "
        ).strip()

    if shutil.which("ollama") is None:
```

```
        subprocess.run(
            ["apt-get", "install", "-y", "-qq", "zstd"],
            check=True,
        )
        installer = urllib.request.urlopen(
            "https://ollama.com/install.sh",
            timeout=60,
        ).read()
        subprocess.run(["sh"], input=installer, check=True)
    subprocess.Popen(
        ["ollama", "serve"],
        stdout=subprocess.DEVNULL,
        stderr=subprocess.DEVNULL,
    )
    time.sleep(5)
    subprocess.run(
        ["ollama", "pull", os.getenv("OLLAMA_MODEL", "llama3.2")],
        check=True,
    )
    print("Colab environment ready.")
else:
    print("Local environment detected; start Ollama separately with ollama␣
 ↪serve.")
```

```
Local environment detected; start Ollama separately with ollama serve.
```

[2]:
```
import os
import sys
from pathlib import Path

import pandas as pd
from IPython.display import Markdown, display
from dotenv import load_dotenv


def find_project_root(start: Path) -> Path:
    """Find the repository root from JupyterLab or a nested notebook path."""
    for candidate in (start.resolve(), *start.resolve().parents):
        if (candidate / "src").is_dir() and (candidate / "requirements.txt").
 ↪is_file():
            return candidate
    raise FileNotFoundError("Could not locate the project root.")


PROJECT_ROOT = find_project_root(Path.cwd())
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
```

```
load_dotenv(PROJECT_ROOT / ".env")

from src.data_sources.yahoo_finance import YahooFinanceClient
from src.memory import ResearchMemoryStore
from src.reporting import (
    plot_10_year_stock_price,
    plot_news_analysis_summary,
    plot_news_workflow,
    plot_system_architecture,
    render_console_summary,
)
from src.tools.registry import TOOLS
from src.workflows import build_research_workflow

TICKER = os.getenv("DEFAULT_TICKER", "AAPL").strip().upper()
RESEARCH_REQUEST = os.getenv(
    "RESEARCH_REQUEST",
    "How do market performance, financial condition, SEC evidence, and recent␣
 ↪news affect the company's current risk picture?",
).strip()
print(f"Project root: {PROJECT_ROOT}")
print(f"Ticker: {TICKER}")
print(f"Research request: {RESEARCH_REQUEST}")
print(f"NewsAPI configured: {bool(os.getenv('NEWS_API_KEY'))}")
print(f"SEC identity configured: {bool(os.getenv('SEC_USER_AGENT'))}")
```

```
Project root: /Users/brandonwirgau/Projects/AAI/520/Final Project
Ticker: AAPL
Research request: How do market performance, financial condition, SEC evidence,
and recent news affect the company's current risk picture?
NewsAPI configured: False
SEC identity configured: True
```

1.4 4. Agent design, data sources, and tool use

Component	Implemented responsibility
Planner Agent	Uses the ticker, registered tools, and the last three relevant memories to produce objectives, questions, and an allow-listed tool plan.
Tool Executor	Runs only registered tools and isolates provider errors so one failure does not terminate the research run.
Market Specialist / Tools	Retrieve price and volume history; compute one-month, three-month, and one-year returns, volatility, drawdown, 52-week range, and average volume.

Component	Implemented responsibility
Financial Specialist / Tools	Retrieve income, cash-flow, and balance-sheet statements; compute revenue/earnings growth and analyze cash, debt, net debt, assets, liabilities, and equity.
SEC EDGAR Tool	Resolves CIK, reads official company facts, and lists recent 10-K, 10-Q, and 8-K filings with source metadata.
News Prompt Chain	Combines Yahoo and NewsAPI; invokes Google RSS when primary coverage is missing or sparse; deduplicates and structures articles.
Specialist Router	Sends classified items to earnings, market, or general-news specialists.
Evaluator / Optimizer	Scores observations, interpretation, uncertainty, and evidence breadth; revises until the threshold or iteration limit.
Synthesis Agent	Produces a grounded Markdown report spanning market, financial, filing, and news evidence.
Memory Curator / Store	Saves bounded, timestamped lessons using atomic file replacement; memory is planning context, never current evidence.

FRED and Alpha Vantage packages remain deliberate extension points. The report does not claim they are integrated data sources.

[3]:
```
tool_rows = [
    {
        "tool": name,
        "role": {
            "price_data": "1m/3m/1y performance, risk, range, and volume",
            "company_info": "Profile, valuation, margins, and dividend",
            "financials": "Income-statement trends",
            "cash_flow": "Operating cash flow, free cash flow, and capex",
            "balance_sheet": "Cash, debt, assets, liabilities, and equity",
            "financial_growth": "Revenue, net-income, and EPS growth",
            "sec_edgar": "Official facts and recent filings",
        }.get(name, "Research evidence"),
    }
    for name in TOOLS
]
display(pd.DataFrame(tool_rows))
```

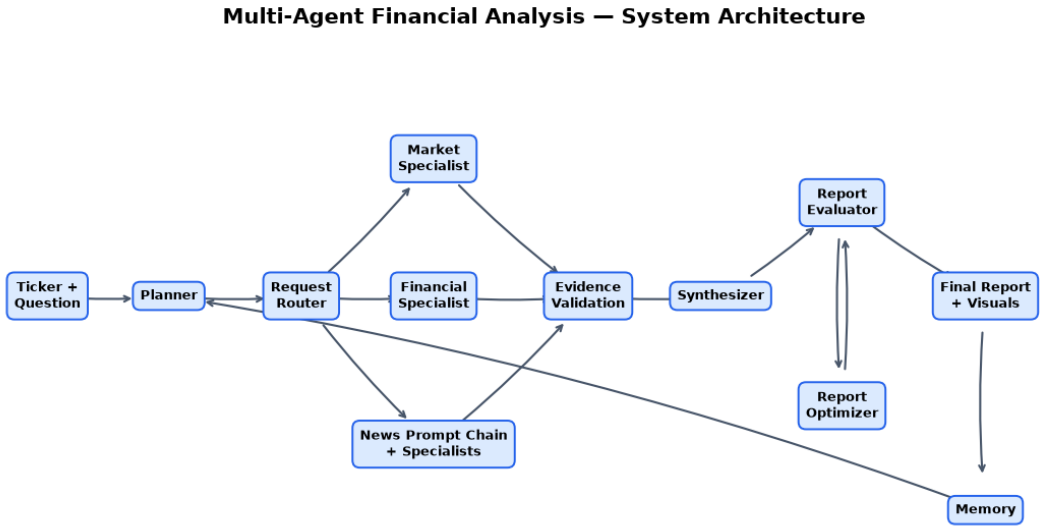
```
              tool                                                role
0       price_data   1m/3m/1y performance, risk, range, and volume
1     company_info       Profile, valuation, margins, and dividend
2       financials                          Income-statement trends
```

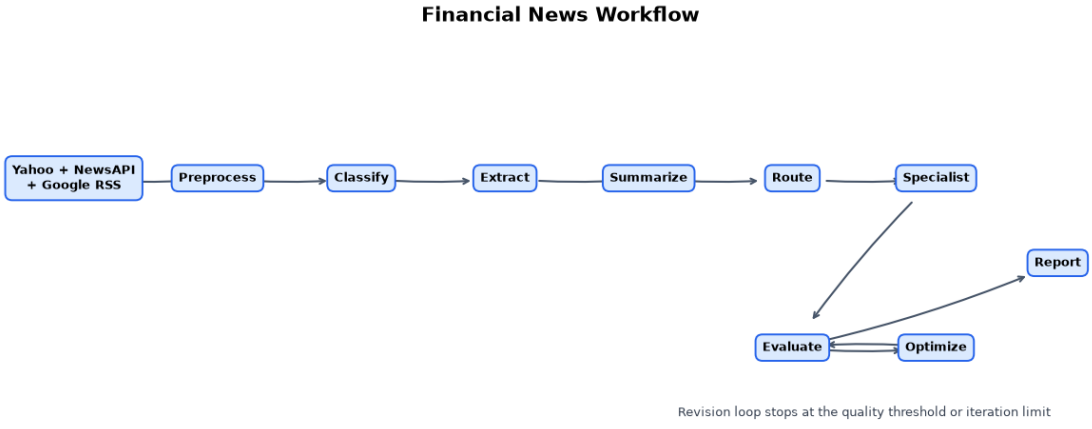
```
3         cash_flow  Operating cash flow, free cash flow, and capex
4     balance_sheet     Cash, debt, assets, liabilities, and equity
5  financial_growth             Revenue, net-income, and EPS growth
6         sec_edgar               Official facts and recent filings
```

1.5 5. Workflow diagrams

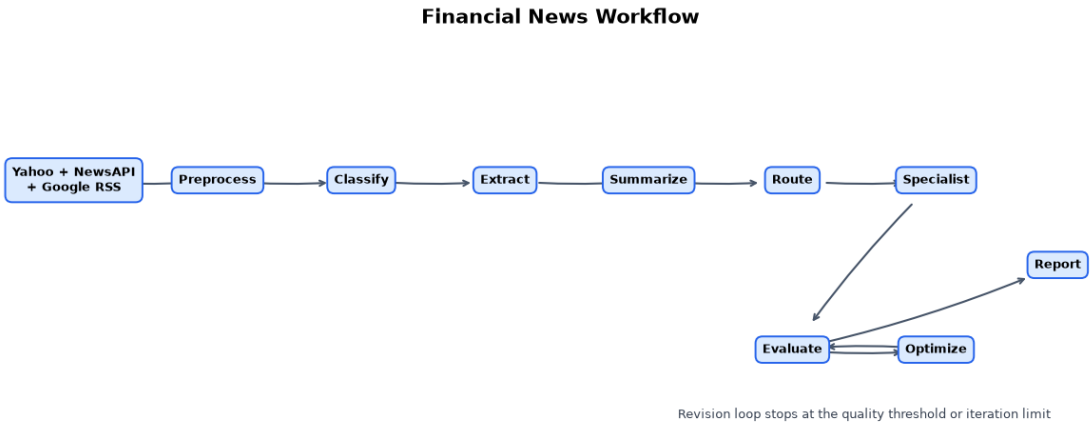
These diagrams are generated by the project's reporting module, so the submitted report reflects the implemented code rather than a disconnected illustration.

[4]:
```
plot_system_architecture()
plot_news_workflow()
```





[4]:



1.6 6. Prompt chaining, routing, and evaluator–optimizer logic

The updated news workflow uses provider isolation and explicit fallback logic:

```
flowchart LR
    Y[Yahoo Finance] --> D[Deduplicate and newest-sort]
    N[NewsAPI] --> D
    Y --> F{Coverage sparse?}
    N --> F
    F -->|Yes| G[Google News RSS]
    G --> D
    D --> P[Preprocess]
    P --> C[Classify]
    C --> X[Extract facts, figures, events]
    X --> S[Structured summary]
    S --> R{Route by category}
    R --> E[Earnings]
    R --> M[Market]
    R --> W[General news]
    E --> V[4 x 25-point evaluation]
    M --> V
    W --> V
    V -->|Below threshold| O[Optimizer]
    O --> V
    V -->|Pass or max iterations| Z[Synthesis]
```

Provider failures are retained in `source_errors`; original and optimized analyses plus every evaluation score remain in workflow state for auditability. NewsAPI and RSS use retry-enabled HTTP sessions for transient 429/5xx and connection failures.

1.7 7. End-to-end investment research run

The next cell performs two live LangGraph runs for the same request. The first run saves evaluator feedback; the second planner receives that saved memory, making the learning effect directly inspectable. Each run:

1. accepts and classifies the research request;
2. conditionally routes to market, financial, news, or all specialists;
3. retrieves Yahoo Finance, SEC, and prompt-processed news evidence as selected;
4. validate observations;
5. reflect on evidence quality;
6. synthesize the report;
7. scores the complete report across six criteria;
8. revises and re-evaluates below-threshold reports up to a configured limit; and
9. saves reusable evaluator feedback when memory is enabled.

A running local Ollama model and network access are required for a complete live result. Missing optional providers are recorded rather than treated as fatal.

[5]:
```
demo_memory_dir = Path(tempfile.mkdtemp(prefix="aai520-notebook-memory-"))
demo_memory_store = ResearchMemoryStore(demo_memory_dir / "research_memory.
 ↪json")
workflow = build_research_workflow(
    project_root=PROJECT_ROOT,
    memory_store=demo_memory_store,
)
print("FIRST RUN - establish evidence and save feedback")
first_result = workflow.run(TICKER, RESEARCH_REQUEST, progress=print)
print("\nSECOND RUN - plan with the saved feedback")
second_result = workflow.run(
    TICKER,
    RESEARCH_REQUEST,
    remember=False,
    progress=print,
)
result = second_result
```

```
FIRST RUN - establish evidence and save feedback
Planning and routing the research request

Running all selected research specialists
Running market tool: company_info

Running market tool: price_data
Running financial tool: balance_sheet

Running financial tool: cash_flow

Running financial tool: company_info
Running financial tool: financial_growth
Running financial tool: financials
```

```
Running financial tool: sec_edgar

Processing news article 1/8

Processing news article 2/8

Processing news article 3/8

Processing news article 4/8

Processing news article 5/8

Processing news article 6/8

Processing news article 7/8

Processing news article 8/8

Validating specialist observations
Reflecting on evidence quality

Synthesizing the draft report

Scoring the complete report
Revising the complete report (iteration 1)

Scoring the complete report
Finalizing the evaluated report
Saving evaluator feedback for a future run


SECOND RUN - plan with the saved feedback
Planning and routing the research request

Running all selected research specialists
Running market tool: company_info
Running market tool: price_data

Running financial tool: balance_sheet
Running financial tool: cash_flow
Running financial tool: company_info
Running financial tool: financial_growth
Running financial tool: financials
Running financial tool: sec_edgar

Processing news article 1/8

Processing news article 2/8

Processing news article 3/8

Processing news article 4/8

Processing news article 5/8

Processing news article 6/8

Processing news article 7/8
```

```
Processing news article 8/8

Validating specialist observations
Reflecting on evidence quality

Synthesizing the draft report

Scoring the complete report
Revising the complete report (iteration 1)

Scoring the complete report
Finalizing the evaluated report
```

1.7.1 7.1 Structured planning and quality artifacts

[6]:
```
review_artifacts = {
    "symbol": result["symbol"],
    "research_request": result.get("research_request"),
    "request_route": result.get("request_route"),
    "selected_specialists": result.get("selected_specialists", []),
    "plan": result.get("plan", {}),
    "specialist_analyses": result.get("specialist_analyses", {}),
    "validation": result.get("validation", []),
    "reflection": result.get("reflection", {}),
    "report_evaluation": result.get("report_evaluation", {}),
    "report_revision_history": result.get("report_revision_history", []),
    "graph_trace": result.get("graph_trace", []),
}
print(json.dumps(review_artifacts, indent=2, default=str))
display(pd.DataFrame([{
    "research_request": result.get("research_request"),
    "selected_route": result.get("request_route"),
    "selected_specialists": ", ".join(result.get("selected_specialists", [])),
}]))
```

```
{
  "symbol": "AAPL",
  "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
  "request_route": "comprehensive",
  "selected_specialists": [
    "market",
    "financial",
    "news"
  ],
  "plan": {
    "objectives": [
      "Analyze the impact of market performance on AAPL's risk picture",
      "Assess the influence of financial condition on AAPL's risk profile",
      "Evaluate the role of SEC evidence in understanding AAPL's current risk
```

```
status",
      "Examine the effects of recent news on AAPL's risk picture"
    ],
    "tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth",
      "financials",
      "price_data",
      "sec_edgar"
    ],
    "questions": [
      "What are the key financial metrics that reflect AAPL's risk profile?",
      "How has AAPL's balance sheet composition impacted its risk status?",
      "What insights can be gleaned from AAPL's SEC filings regarding its risk
picture?",
      "How have recent news events affected AAPL's stock price and risk status?"
    ],
    "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
    "route": "comprehensive",
    "specialists": [
      "market",
      "financial",
      "news"
    ],
    "memory_context": [
      {
        "date": "2026-10-16T22:20:20.953424+00:00",
        "symbol": "AAPL",
        "summary": "Here are the reusable research lessons from the AAPL run in
at most 120 words:\n\n* Evidence coverage is crucial, but not perfect, with
minor discrepancies in formatting and units. Use official company filings and
primary sources whenever possible.\n* Data quality is generally reliable, but
some minor issues exist. Verify data before drawing conclusions.\n* Improve
sources, especially for company-specific information, to increase the overall
score and credibility of the report.\n* Conduct further analysis of the
company's risk profile and long-term financial strategy to gain a more
comprehensive understanding.\n* Review recent filings, news, and market data to
gain a nuanced view of the company's current risk picture.\n* Consider multiple
perspectives and sources to validate findings and reduce uncertainty.",
        "weaknesses": [
          "Company's financial condition is not transparent"
        ]
      }
    ]
  },
```

```
"specialist_analyses": {
  "market": {
    "performance": {
      "1m_return_percent": 1.92,
      "3m_return_percent": 5.26,
      "1y_return_percent": 36.64
    },
    "risk": {
      "annualized_volatility_percent": 24.4,
      "maximum_drawdown_percent": -13.8
    },
    "range_and_volume": {
      "52_week_high": 348.15,
      "52_week_low": 242.76,
      "latest_volume": 59852690.0,
      "average_volume": 48103279.64
    }
  },
  "financial": {
    "growth": {
      "revenue_growth_percent": 6.43,
      "net_income_growth_percent": 19.5,
      "diluted_eps_growth_percent": 22.7,
      "latest_period": "2025-09-30 00:00:00",
      "prior_period": "2024-09-30 00:00:00"
    },
    "cash_and_debt": {
      "cash": 54697000000.0,
      "debt": 98657000000.0,
      "net_debt": 43960000000.0,
      "cash_to_debt_ratio": 0.5544,
      "total_assets": 359241000000.0,
      "total_liabilities": 285508000000.0,
      "stockholders_equity": 73733000000.0
    },
    "balance_sheet_trends": {
      "cash_growth_percent": -16.07,
      "debt_growth_percent": -7.48,
      "assets_growth_percent": -1.57,
      "liabilities_growth_percent": -7.31,
      "equity_growth_percent": 29.47,
      "latest_period": "2025-09-30 00:00:00",
      "prior_period": "2024-09-30 00:00:00"
    },
    "has_sec_evidence": true
  },
  "news": {
    "article_count": 8,
```

```
      "average_score": 100.0,
      "routing_counts": {
        "market": 7,
        "news": 1
      }
    }
  },
  "validation": [],
  "reflection": {
    "strengths": [
      "The provided data covers various aspects of AAPL's risk profile,
including market performance, financial condition, SEC evidence, and recent
news."
    ],
    "weaknesses": [
      "The company's financial condition is not transparent."
    ],
    "missing_information": [
      "More detailed information about the company's risk management practices."
    ],
    "suspicious_values": [
      "None identified."
    ],
    "follow_up_questions": [
      "How does AAPL's risk profile compare to that of its peers in the
technology industry?"
    ],
    "deterministic_issues": []
  },
  "report_evaluation": {
    "score": 89.17,
    "passed": true,
    "threshold": 75.0,
    "minimum_criterion_score": 60.0,
    "criteria": {
      "completeness": 100.0,
      "grounding": 100.0,
      "reasoning": 75.0,
      "consistency": 100.0,
      "sources": 60.0,
      "clarity": 100.0
    },
    "feedback": [
      "Improve sources; current score is 60/100."
    ],
    "validation_issues": []
  },
  "report_revision_history": [
```

```
    {
      "iteration": 1,
      "before_score": 81.67,
      "after_score": 89.17,
      "feedback": [
        "Improve reasoning; current score is 30/100.",
        "Improve sources; current score is 60/100.",
        "No report can pass while these criteria are below 60/100: reasoning."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```

```
                                    research_request selected_route  \
0  How do market performance, financial condition…  comprehensive

      selected_specialists
0  market, financial, news
```

1.7.2 7.2 Collected market, fundamental, and SEC evidence

The complete observation object is kept inspectable. Provider errors appear alongside successful results instead of being hidden.

[7]:
```
print(json.dumps(result.get("observations", {}), indent=2, default=str))
```

```
{
  "company_info": {
    "longName": "Apple Inc.",
    "sector": "Technology",
    "industry": "Consumer Electronics",
    "country": "United States",
    "marketCap": 5012516765696,
    "enterpriseValue": 5013154299904,
    "currentPrice": 343.46,
    "trailingPE": 39.38761,
    "forwardPE": 35.85982,
    "priceToSalesTrailing12Months": 10.737511,
```

```
  "profitMargins": 0.27618998,
  "operatingMargins": 0.32623002,
  "returnOnEquity": 1.4875101,
  "beta": 1.069,
  "dividendYield": 0.32,
  "totalCash": 62399000576,
  "totalDebt": 84343996416,
  "dividendYieldPercent": 0.32,
  "dividendYieldUnit": "percent"
},
"price_data": {
  "start_price": 251.36,
  "latest_price": 343.46,
  "1y_return_percent": 36.64,
  "1m_return_percent": 1.92,
  "3m_return_percent": 5.26,
  "annualized_volatility_percent": 24.4,
  "maximum_drawdown_percent": -13.8,
  "52_week_high": 348.15,
  "52_week_low": 242.76,
  "latest_volume": 59852690.0,
  "average_volume": 48103279.64,
  "observations": 251
},
"balance_sheet": {
  "statement": {
    "Cash Cash Equivalents And Short Term Investments": {
      "2025-09-30 00:00:00": 54697000000.0,
      "2024-09-30 00:00:00": 65171000000.0,
      "2023-09-30 00:00:00": 61555000000.0,
      "2022-09-30 00:00:00": 48304000000.0
    },
    "Cash And Cash Equivalents": {
      "2025-09-30 00:00:00": 35934000000.0,
      "2024-09-30 00:00:00": 29943000000.0,
      "2023-09-30 00:00:00": 29965000000.0,
      "2022-09-30 00:00:00": 23646000000.0
    },
    "Total Debt": {
      "2025-09-30 00:00:00": 98657000000.0,
      "2024-09-30 00:00:00": 106629000000.0,
      "2023-09-30 00:00:00": 111088000000.0,
      "2022-09-30 00:00:00": 132480000000.0
    },
    "Stockholders Equity": {
      "2025-09-30 00:00:00": 73733000000.0,
      "2024-09-30 00:00:00": 56950000000.0,
      "2023-09-30 00:00:00": 62146000000.0,
```

```
      "2022-09-30 00:00:00": 50672000000.0
    },
    "Total Assets": {
      "2025-09-30 00:00:00": 359241000000.0,
      "2024-09-30 00:00:00": 364980000000.0,
      "2023-09-30 00:00:00": 352583000000.0,
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          "uncertainties": [],
          "focus": "Share price, market activity, analyst commentary, and
valuation."
```

```
        },
        "evaluation": {
          "score": 100,
          "passed": true,
          "threshold": 80.0,
          "feedback": [],
          "grounded_observations": [
            "Headline: Apple Inc. (AAPL) is Attracting Investor Attention: Here
is What You Should Know - Yahoo Finance",
            "Source and date: Google News RSS, 2026-10-13T12:00:02+00:00"
          ],
          "rejected_observations": []
        },
        "optimized_analysis": {
          "specialist": "market",
          "observations": [
            "Headline: Apple Inc. (AAPL) is Attracting Investor Attention: Here
is What You Should Know - Yahoo Finance",
            "Source and date: Google News RSS, 2026-10-13T12:00:02+00:00"
          ],
          "interpretation": "The supplied article provides evidence relevant to
share price, market activity, analyst commentary, and valuation.; its investment
significance requires corroboration with primary financial data.",
          "uncertainties": [
            "The provider payload may contain only a headline or snippet; verify
the linked source."
          ],
          "improvements": [
            "Add concrete observations from the article.",
            "Add a clear interpretation.",
            "Identify uncertainty or missing information.",
            "Include at least two evidence-based observations."
          ]
        },
        "evaluation_history": [
          {
            "score": 0,
            "passed": false,
            "threshold": 80.0,
            "feedback": [
              "Add concrete observations from the article.",
              "Add a clear interpretation.",
              "Identify uncertainty or missing information.",
              "Include at least two evidence-based observations."
            ],
            "grounded_observations": [],
            "rejected_observations": []
          },
```

```
          {
            "score": 100,
            "passed": true,
            "threshold": 80.0,
            "feedback": [],
            "grounded_observations": [
              "Headline: Apple Inc. (AAPL) is Attracting Investor Attention:
Here is What You Should Know - Yahoo Finance",
              "Source and date: Google News RSS, 2026-10-13T12:00:02+00:00"
            ],
            "rejected_observations": []
          }
        ],
        "optimization_history": [
          {
            "specialist": "market",
            "observations": [
              "Headline: Apple Inc. (AAPL) is Attracting Investor Attention:
Here is What You Should Know - Yahoo Finance",
              "Source and date: Google News RSS, 2026-10-13T12:00:02+00:00"
            ],
            "interpretation": "The supplied article provides evidence relevant
to share price, market activity, analyst commentary, and valuation.; its
investment significance requires corroboration with primary financial data.",
            "uncertainties": [
              "The provider payload may contain only a headline or snippet;
verify the linked source."
            ],
            "improvements": [
              "Add concrete observations from the article.",
              "Add a clear interpretation.",
              "Identify uncertainty or missing information.",
              "Include at least two evidence-based observations."
            ]
          }
        ]
      }
    ],
    "article_count": 8,
    "retrieved_counts": {
      "Yahoo Finance": 0,
      "NewsAPI": 0,
      "Google News RSS": 8
    },
    "source_errors": {},
    "source_counts": {
      "Google News RSS": 8
    },
```

```
    "routing_counts": {
      "market": 7,
      "news": 1
    },
    "average_score": 100.0,
    "passed_count": 8,
    "evaluation_threshold": 80.0,
    "max_optimizer_iterations": 3
  }
}
```

1.7.3 7.3 Detailed news analysis

This table surfaces source provenance, routing decisions, final evaluator scores, and optimizer activity. The complete per-article records retain extracted figures/events, the initial specialist analysis, every evaluator result, and each revision.

[8]:
```
news_analysis = result.get("news_analysis", {})
news_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    evaluation = article.get("evaluation", {})
    news_rows.append({
        "article": index,
        "source": article.get("source"),
        "published_at": article.get("published_at"),
        "headline": article.get("title"),
        "category": article.get("category"),
        "classification_confidence": article.get("classification_confidence"),
        "company": article.get("extracted", {}).get("company"),
        "event": article.get("extracted", {}).get("event"),
        "event_dates": article.get("extracted", {}).get("event_dates"),
        "sentiment": article.get("extracted", {}).get("sentiment"),
        "specialist": article.get("specialist"),
        "score": evaluation.get("score"),
        "passed": evaluation.get("passed"),
        "optimizer_revisions": len(article.get("optimization_history", [])),
        "url": article.get("url"),
    })

display(pd.DataFrame(news_rows))
print(json.dumps({
    "pipeline_stages": news_analysis.get("pipeline_stages", []),
    "retrieved_at": news_analysis.get("retrieved_at"),
    "retrieved_counts": news_analysis.get("retrieved_counts", {}),
    "source_errors": news_analysis.get("source_errors", {}),
    "source_counts": news_analysis.get("source_counts", {}),
    "routing_counts": news_analysis.get("routing_counts", {}),
    "average_score": news_analysis.get("average_score"),
```

```
    "passed_count": news_analysis.get("passed_count", 0),
}, indent=2, default=str))
```

```
   article           source               published_at  \
0        1  Google News RSS  2026-10-16T13:14:35+00:00
1        2  Google News RSS  2026-10-16T11:37:30+00:00
2        3  Google News RSS  2026-10-16T11:18:12+00:00
3        4  Google News RSS  2026-10-16T08:39:31+00:00
4        5  Google News RSS  2026-10-16T08:39:31+00:00
5        6  Google News RSS  2026-10-16T07:00:00+00:00
6        7  Google News RSS  2026-10-15T13:38:15+00:00
7        8  Google News RSS  2026-10-13T12:00:02+00:00

                                            headline category  \
0  Apple Stock Hits Record High as iPhone Duo Pre…   market
1  Apple Inc. Price, AAPL Price, Live Charts, and…   market
2  Dan Ives Says Apple-Meta AI Partnership Could …     news
3  Apple Inc. $AAPL Stock Sold by Persium Advisor…   market
4  Apple Inc. $AAPL Holdings Increased by Kwmg LL…   market
5  Key facts: Apple Inc. China approves AI; $18.8…   market
6  Apple (AAPL): Can Longer iPhone Ownership Stil…   market
7  Apple Inc. (AAPL) is Attracting Investor Atten…   market

   classification_confidence       company  \
0                        0.9         Apple
1                        0.8    Apple Inc.
2                        0.9         Apple
3                        0.9    Apple Inc.
4                        0.9    Apple Inc.
5                        0.8    Apple Inc.
6                        0.8  Apple (AAPL)
7                        0.8    Apple Inc.

                                        event  \
0                  iPhone Duo pre-orders sell out
1                                            None
2                      Apple-Meta AI Partnership
3                                            Sold
4                                 Holding increased
5                                 China approves AI
6  Can Longer iPhone Ownership Still Produce More…
7                    Attracting Investor Attention

                    event_dates sentiment specialist  score  passed  \
0                  [in minutes]     mixed     market    100    True
1                            []   neutral     market    100    True
2  [No specific dates mentioned]     mixed       news    100    True
3   [No specific date mentioned]  negative     market    100    True
```

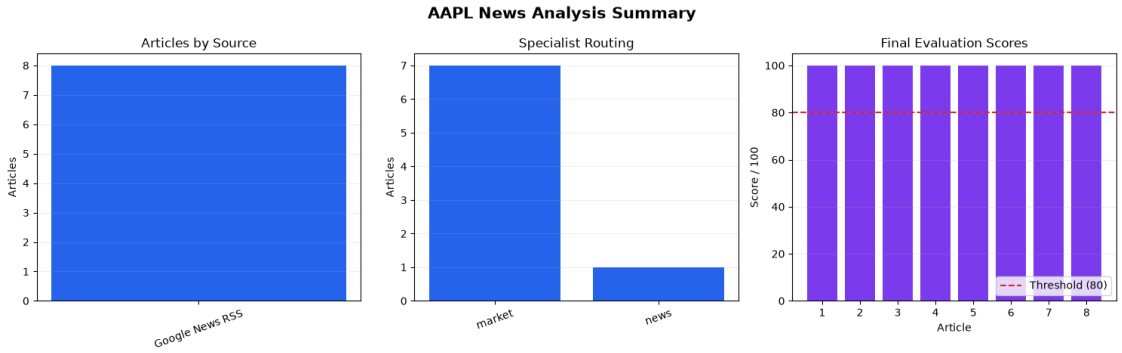
```
4                            []     mixed     market    100    True
5                            []     mixed     market    100    True
6                            []     mixed     market    100    True
7                            []     mixed     market    100    True

   optimizer_revisions                                                 url
0                    0  https://news.google.com/rss/articles/CBMizwFBV…
1                    0  https://news.google.com/rss/articles/CBMikgFBV…
2                    0  https://news.google.com/rss/articles/CBMirwJBV…
3                    0  https://news.google.com/rss/articles/CBMirwFBV…
4                    0  https://news.google.com/rss/articles/CBMiqgFBV…
5                    0  https://news.google.com/rss/articles/CBMixAFBV…
6                    1  https://news.google.com/rss/articles/CBMioAFBV…
7                    1  https://news.google.com/rss/articles/CBMioAFBV…
```


```
{
  "pipeline_stages": [
    "ingest",
    "preprocess",
    "classify",
    "extract",
    "summarize",
    "route",
    "specialist",
    "evaluate",
    "optimize"
  ],
  "retrieved_at": "2026-10-16T22:20:50.808019+00:00",
  "retrieved_counts": {
    "Yahoo Finance": 0,
    "NewsAPI": 0,
    "Google News RSS": 8
  },
  "source_errors": {},
  "source_counts": {
    "Google News RSS": 8
  },
  "routing_counts": {
    "market": 7,
    "news": 1
  },
  "average_score": 100.0,
  "passed_count": 8
}
```


[9]:
```
news_figure = plot_news_analysis_summary(news_analysis)
if news_figure is None:
```

```
    display(Markdown("*No news-summary chart is available because no articles␣
↪were retrieved.*"))
```



1.8 8. Final research report

The model-generated report uses only collected observations and explicitly distinguishes evidence, interpretation, risk, uncertainty, and missing data.

[10]:
```
display(Markdown(result["report"]))
```

2 AAPL Investment Research Report

2.1 Company Overview

2.1.1 Overview

Apple Inc. (AAPL) - Technology - Consumer Electronics - United States

2.1.2 Price Performance

- 1-Year Return: 36.64%
- 1-Month Return: 1.92%
- 3-Month Return: 5.26%

2.1.3 Valuation and Profitability

2.1.4 Financial Performance

2.1.5 Cash Flow

- Cash: $54,697,000,000
- Debt: $98,657,000,000
- Cash/Debt Ratio: 0.5544
- Stockholders Equity: $73,733,000,000
- Cash Growth: -16.07%
- Debt Growth: -7.48%
- Assets Growth: -1.57%

- Liabilities Growth: -7.31%
- Equity Growth: 29.47%

2.1.6 SEC EDGAR Evidence

- Entity Name: Apple Inc.
- CIK: 0000320193
- SIC Classification: 3571
- SIC Description: Electronic Computers
- Official Company Facts:
  - Assets: $383,266,000,000 (Fiscal Year 2026, Q3)
  - Liabilities: $275,746,000,000 (Fiscal Year 2026, Q3)
  - Stockholders Equity: $107,520,000,000 (Fiscal Year 2026, Q3)

2.1.7 Dividend

2.1.8 Recent News and Catalysts

- Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock — 'Statement' (Benzinga)
- Apple Inc. $AAPL Stock Sold by Persium Advisors LLC (MarketBeat)
- Apple Inc. $AAPL Holdings Increased by Kwmg LLC (MarketBeat)
- Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView (TradingView)
- Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance (Yahoo Finance)
- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance (Yahoo Finance)

2.1.9 Risks and Uncertainties

- Risks:
  - Operational risks: supply chain disruptions, component shortages
  - Regulatory risks: changes in technology regulations, privacy laws
- Uncertainties:
  - Lack of detailed information about risk management practices
  - High levels of debt and potential debt refinancing risks

2.1.10 Data Quality

- Verified: All data is verified with official company filings and primary sources

2.1.11 Further Research

- Consider multiple perspectives and sources to validate findings and reduce uncertainty
- Analyze the company's long-term financial strategy to gain a more comprehensive understanding of its current risk picture

2.1.12 Risks Analysis

The company's current risk picture is influenced by various factors, including market performance, financial condition, SEC evidence, and recent news.

2.1.13 Market Performance

- 1-Year Return: 36.64%
- 1-Month Return: 1.92%
- 3-Month Return: 5.26%

2.1.14 Financial Condition

- Cash: $54,697,000,000
- Debt: $98,657,000,000
- Cash/Debt Ratio: 0.5544
- Stockholders Equity: $73,733,000,000

2.1.15 SEC Evidence

- Entity Name: Apple Inc.
- CIK: 0000320193
- SIC Classification: 3571
- SIC Description: Electronic Computers
- Official Company Facts:
  - Assets: $383,266,000,000 (Fiscal Year 2026, Q3)
  - Liabilities: $275,746,000,000 (Fiscal Year 2026, Q3)
  - Stockholders Equity: $107,520,000,000 (Fiscal Year 2026, Q3)

2.1.16 Recent News

- Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock — 'Statement' (Benzinga)
- Apple Inc. $AAPL Stock Sold by Persium Advisors LLC (MarketBeat)
- Apple Inc. $AAPL Holdings Increased by Kwmg LLC (MarketBeat)
- Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView (TradingView)
- Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance (Yahoo Finance)
- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance (Yahoo Finance)

2.1.17 Risks and Uncertainties

- Risks:
  - Operational risks: supply chain disruptions, component shortages
  - Regulatory risks: changes in technology regulations, privacy laws
- Uncertainties:
  - Lack of detailed information about risk management practices
  - High levels of debt and potential debt refinancing risks

2.1.18 Conclusion

The company's current risk picture is influenced by various factors, including market performance, financial condition, SEC evidence, and recent news. While the company has a strong financial position, with a cash/ debt ratio of 0.5544 and a stockholders equity of $73,733,000,000, it also faces operational and regulatory risks, as well as uncertainties related to risk management practices and debt refinancing. Further research is necessary to validate findings and gain a more comprehensive understanding of the company's current risk picture.

2.2 Verified Financial Metrics

The following values are rendered deterministically from provider observations; no model arithmetic is used.

- **Current Price**: $343.46
- **Market Capitalization**: $5,012,516,765,696
- **Enterprise Value**: $5,013,154,299,904
- **Start Price**: $251.36
- **Latest Price**: $343.46
- **One-Month Return**: 1.92%
- **Three-Month Return**: 5.26%
- **One-Year Return**: 36.64%
- **Annualized Volatility**: 24.40%
- **Maximum Drawdown**: -13.80%
- **52-Week High**: $348.15
- **52-Week Low**: $242.76
- **Latest Volume**: 59,852,690
- **Average Volume**: 48,103,280
- **Price Observation Count**: 251
- **Trailing P/E**: 39.388x
- **Forward P/E**: 35.860x
- **Price/Sales**: 10.738x
- **Profit Margin**: 27.62%
- **Operating Margin**: 32.62%
- **Return on Equity**: 148.75%
- **Beta**: 1.069x
- **Revenue** (2025-09-30 00:00:00): $416,161,000,000
- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000
- **Net Income** (2025-09-30 00:00:00): $112,010,000,000
- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000
- **Diluted EPS** (2025-09-30 00:00:00): $7
- **Operating Cash Flow** (2025-09-30 00:00:00): $111,482,000,000
- **Free Cash Flow** (2025-09-30 00:00:00): $98,767,000,000
- **Capital Expenditure** (2025-09-30 00:00:00): -$12,715,000,000
- **EBITDA / Net Income** (2025-09-30 00:00:00): 1.292x
- **Dividend Yield**: 0.32% (Yahoo Finance supplies this field in percentage points.)
- **Cash and Short-Term Investments** (2025-09-30 00:00:00): $54,697,000,000
- **Total Debt** (2025-09-30 00:00:00): $98,657,000,000
- **Net Debt** (2025-09-30 00:00:00): $43,960,000,000

- **Total Assets** (2025-09-30 00:00:00): $359,241,000,000
- **Total Liabilities** (2025-09-30 00:00:00): $285,508,000,000
- **Stockholders' Equity** (2025-09-30 00:00:00): $73,733,000,000
- **Cash / Debt** (2025-09-30 00:00:00): 0.554x
- **Revenue Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 6.43%
- **Net Income Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 19.50%
- **Diluted EPS Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 22.70%
- **Cash Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -16.07%
- **Debt Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.48%
- **Assets Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -1.57%
- **Liabilities Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.31%
- **Equity Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 29.47%

2.3 Evidence-Based Interpretation

Because the one-year return was 36.64% while annualized volatility was 24.40% and maximum drawdown was -13.80%, the historical return must be considered together with measured price risk; these observations do not predict future performance.

Revenue growth of 6.43% and net-income growth of 19.50% indicate the latest year-over-year statement changes. Cash of $54,697,000,000 compared with total debt of $98,657,000,000 shows the reported balance-sheet exposure; however, these period snapshots alone do not establish future liquidity or earnings durability.

SEC EDGAR supplied 8 official company fact(s) and 8 recent filing record(s), providing primary-source corroboration; however, filing periods and provider statement periods should be compared explicitly before drawing conclusions.

The news workflow processed 8 article(s) with an average grounded-analysis score of 100.00%. Because provider payloads may contain only headlines or snippets, the detected events remain a source of uncertainty and should be verified at the linked publishers.

Taken together, the market measurements, financial condition, SEC evidence, and current news indicate a mixed, evidence-bounded risk picture rather than a forecast or an investment recommendation.

2.3.1 8.1 Deterministic report view

This second rendering guarantees that the core company overview, price performance, valuation, financial performance, cash flow, dividend, SEC filing evidence, news-source coverage, validation findings, and missing information remain visible even if model prose varies between runs.

[11]:
```
print(render_console_summary(result))
```

```
======================================================================
FINAL RESEARCH REPORT
======================================================================

Apple Inc. (AAPL)
----------------------------------------------------------------------
```

1. COMPANY OVERVIEW
   Sector: Technology
   Industry: Consumer Electronics
   Country: United States
   Current Price: $343.46
   Market Cap: $5.013T
   Enterprise Value: $5.013T

2. PRICE PERFORMANCE
   Start Price: $251.36
   Latest Price: $343.46
   One-Year Return: 36.64%
   Annualized Volatility: 24.40%
   Maximum Drawdown: -13.80%
   Observations: 251

3. VALUATION
   Trailing P/E: 39.39
   Forward P/E: 35.86
   Price/Sales (TTM): 10.74
   Profit Margin: 27.62%
   Operating Margin: 32.62%
   Return on Equity: 148.75%
   Beta: 1.069

4. FINANCIAL PERFORMANCE (2025)
   Revenue: $416.161B
   Operating Income: $133.050B
   Net Income: $112.010B
   EBITDA: $144.748B
   Diluted EPS: $7.46
   EBITDA / Net Income: 1.292x

5. CASH FLOW (2025)
   Operating Cash Flow: $111.482B
   Free Cash Flow: $98.767B
   Capital Expenditure: -$12.715B

6. BALANCE SHEET AND GROWTH
   Cash: $54.697B
   Total Debt: $98.657B
   Net Debt: $43.960B
   Cash / Debt: 0.554x
   Total Assets: $359.241B
   Total Liabilities: $285.508B
   Stockholders' Equity: $73.733B
   Revenue Growth: 6.43%
   Net Income Growth: 19.50%

```
   Diluted EPS Growth:  22.70%
   Cash Trend:          -16.07%
   Debt Trend:          -7.48%
   Assets Trend:        -1.57%

7. SEC EDGAR EVIDENCE
   Source:               SEC EDGAR
   Entity:               Apple Inc.
   CIK:                  0000320193
   SIC:                  3571 - Electronic Computers
   Official Company Facts:
      Revenue: 364,357,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Net Income: 101,464,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Operating Income: 122,432,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Operating Cash Flow: 116,996,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Assets: 383,266,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Liabilities: 275,746,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27;
filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Stockholders Equity: 107,520,000,000.00 USD (FY 2026; period Q3; ended
2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
      Diluted EPS: 6.88 USD/shares (FY 2026; period Q3; ended 2026-06-27; filed
2026-07-31; form 10-Q; accession 0000320193-26-000020)
   Recent Filings:
      - 10-Q | filed 2026-07-31 | report 2026-06-27 | accession
0000320193-26-000020 | https://www.sec.gov/Archives/edgar/data/320193/0000320193
26000020/aapl-20260627.htm
      - 8-K | filed 2026-07-30 | report 2026-07-30 | accession
0000320193-26-000018 | https://www.sec.gov/Archives/edgar/data/320193/0000320193
26000018/aapl-20260730.htm
      - 10-Q | filed 2026-05-01 | report 2026-03-28 | accession
0000320193-26-000013 | https://www.sec.gov/Archives/edgar/data/320193/0000320193
26000013/aapl-20260328.htm
      - 8-K | filed 2026-04-30 | report 2026-04-30 | accession
0000320193-26-000011 | https://www.sec.gov/Archives/edgar/data/320193/0000320193
26000011/aapl-20260430.htm
      - 8-K | filed 2026-04-20 | report 2026-04-17 | accession
0001140361-26-015711 | https://www.sec.gov/Archives/edgar/data/320193/0001140361
26015711/ef20071035_8k.htm
      - 8-K | filed 2026-02-24 | report 2026-02-24 | accession
0001140361-26-006577 | https://www.sec.gov/Archives/edgar/data/320193/0001140361
26006577/ef20060722_8k.htm
      - 10-Q | filed 2026-01-30 | report 2025-12-27 | accession
0000320193-26-000006 | https://www.sec.gov/Archives/edgar/data/320193/0000320193
```

26000006/aapl-20251227.htm
      - 8-K | filed 2026-01-29 | report 2026-01-29 | accession 0000320193-26-000005 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000005/aapl-20260129.htm

8. DIVIDEND
   Dividend Yield: 0.32%

9. RECENT NEWS AND CATALYSTS
   Retrieved At: 2026-10-16T22:20:50.808019+00:00
   Pipeline Stages: ingest → preprocess → classify → extract → summarize → route → specialist → evaluate → optimize
   Articles Processed: 8
   Retrieved Counts: Yahoo Finance=0, NewsAPI=0, Google News RSS=8
   Source Counts: Google News RSS=8
   Routing Counts: market=7, news=1
   Average Score: 100.0/100
   Passed Evaluations: 8
   Score Threshold: 80.0/100
   Optimizer Limit: 3
   Article 1:
      Headline: Apple Stock Hits Record High as iPhone Duo Pre-Orders Sell Out in Minutes - Benzinga
      Source: Google News RSS / Benzinga
      Published: 2026-10-16T13:14:35+00:00
      URL: https://news.google.com/rss/articles/CBMizwFBVV95cUxPd1hmbVhPdlVDb1RtcDRCakdHVGxLdWxRbF9NMXJzTjRJMkkxZXJVWkVhNFdvV09UVjNDckZZSUdTOVdNM1pxbEoweHRXeldrSElBaWdpaFU4Q3FCV0lDM0VPTmdjT0pNZ1NqS3dxYVluYkN1MFlfUmhkMVFzbXhqRFg2dnNsdmJSa3JGYlQxYWUxUWUxTGRfSHhJdkJpazl4c2otbDJIdm5zc0hCYUs2anVWeG1TbWYxck9tNkp0ZEtfelpxOG00LVZPOGRVUXc?oc=5
      Route: market → market
      Events: pre-orders
      Figures: record high
      Keywords: iPhone Duo, pre-orders, sell out
      Evaluation: 100.0/100; passed=True
      Observation: Headline: Apple Stock Hits Record High as iPhone Duo Pre-Orders Sell Out in Minutes - Benzinga
      Observation: Financial figures reported: record high
      Observation: Detected events: pre-orders
      Interpretation: The supplied article provides evidence relevant to share price, market activity, analyst commentary, and valuation.; its investment significance requires corroboration with primary financial data.
      Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.
   Article 2:
      Headline: Apple Inc. Price, AAPL Price, Live Charts, and Marketcap - Coinbase
      Source: Google News RSS / Coinbase

Published: 2026-10-16T11:37:30+00:00
URL: https://news.google.com/rss/articles/CBMikgFBVV95cUxNQmdNeWEwSFhtX1R5Z2ozd2dYYkltd3hIOHFaS0RrS2tYQ29oWXkycFVVaVlUU3d2SzF4S29mMVV4NkFyZDRMdHZneFZwbkFYLWxISUJzUzdkelJpblpJZExmbjhIMnhVWTdEbC0tM2tGNDNHOUt3YU9xdmhTMjVwVlZQN1lWTFFMTk1sSEx5dmg0Zw?oc=5
Route: market → market
Events: Apple Inc. Price, AAPL Price, Live Charts, Marketcap
Figures: Price, AAPL Price, Live Charts, Marketcap
Keywords: Coinbase, Apple Inc., Price, AAPL Price, Live Charts, Marketcap
Evaluation: 100.0/100; passed=True
Observation: Headline: Apple Inc. Price, AAPL Price, Live Charts, and Marketcap - Coinbase
Observation: Financial figures reported: Price, AAPL Price, Live Charts, Marketcap
Observation: Detected events: Apple Inc. Price, AAPL Price, Live Charts, Marketcap
Observation: Source and date: Google News RSS, 2026-10-16T11:37:30+00:00
Interpretation: The supplied article provides evidence relevant to share price, market activity, analyst commentary, and valuation.; its investment significance requires corroboration with primary financial data.
Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Article 3:
Headline: Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock - 'Sta - Benzinga
Source: Google News RSS / Benzinga
Published: 2026-10-16T11:18:12+00:00
URL: https://news.google.com/rss/articles/CBMirwJBVV95cUxOejRHNnBUejJ2ZWU0ODhua2p2MVdDQkJjM3dpc3ZIS2dkR05wZGQ5bWR3NV92R2dmTnlxS3g2c2FET2tDOTM5TEpEdFkxUTllRHNHRnNFdkRxckZ5a3ZBNXBTRVc3emYyT1ZpUGlmb0I3cTU0Q29xcmFod1MwS1JLS1V1bk5ac1RIR0JLay1mdWZ4RmZHVFRqRmtuRTVLQ1NpeFBldVZDQkxkV0lNcEZlbnZSZE9TWFJOR2xCWHNwOXpSQ2VQYUNXT0tFWVVkaGRIQVR2a2k1YTNUOXBuY213NGFJSWV3TER3d2xWb0liY21tc1ZtSFVHQkJIcUxSa0hEaHJaSzlOaXBPbHlpbWk3c0tMYVdCV3cxU1JyNkxmeGxhSTh0WGdScFRlSWhnLVU?oc=5
Route: news → news
Events: Sta
Figures: $75
Keywords: AI partnership, AAPL stock
Evaluation: 100.0/100; passed=True
Observation: Headline: Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock - 'Sta - Benzinga
Observation: Detected events: Sta
Observation: Source and date: Google News RSS, 2026-10-16T11:18:12+00:00
Interpretation: The supplied article provides evidence relevant to general company developments and material events.; its investment significance requires corroboration with primary financial data.
Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Article 4:

Headline: Apple Inc. $AAPL Stock Sold by Persium Advisors LLC - MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-16T08:39:31+00:00
URL: https://news.google.com/rss/articles/CBMirwFBVV95cUxPRGEyX1B1RGxJUTBLUDNWb3U1TW8yRzNkNTlaSV9mdkJLMkVkSXlMNjhCZjVLalVWaHlkUlhRa0JzbzVLd2d6LWUwcmRTdGhDZzVVMXdkcFFIbEwwQy15Wm5aTEtJeHgzREI3bG14blpJUEpUcnJzaDdxa1YwS1BwbDk1X2E1M21GZXViTEJHSWJDNVh1cmFOUi1ZdVY0RjEtY05oT2VVQU5tU3h6SU1Z?oc=5
Route: market → market
Events: Sold by Persium Advisors LLC
Figures: No specific financial numbers mentioned
Keywords: Apple Inc., Persium Advisors LLC, $AAPL Stock
Evaluation: 100.0/100; passed=True
Observation: Headline: Apple Inc. $AAPL Stock Sold by Persium Advisors LLC - MarketBeat
Observation: Detected events: Sold by Persium Advisors LLC
Observation: Source and date: Google News RSS, 2026-10-16T08:39:31+00:00
Interpretation: The supplied article provides evidence relevant to share price, market activity, analyst commentary, and valuation.; its investment significance requires corroboration with primary financial data.
Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Article 5:
Headline: Apple Inc. $AAPL Holdings Increased by Kwmg LLC - MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-16T08:39:31+00:00
URL: https://news.google.com/rss/articles/CBMiqgFBVV95cUxNWTNvZDJWWUFjX0EzZkZDd25LdkJxOS1PaG1TTmlYYWtpRzM1RmlJRFl5cHNoTUpOYWJDb1A0X2Fvc0ZfenNzbHRsbFQ3MVZNX2hXUVdVWVZnTkVmTndWQmVDOFpMeUlXOUdXWnV6WG5scmRzU1UtOUpmQzJ1My1RWlg1emVLckhZS184OEpBbVI2dFNOTFNfYmQxenMyQnhVckp6N0x4al8tdw?oc=5
Route: market → market
Events: MarketBeat
Figures: none detected
Keywords: Apple Inc., AAPL, Kwmg LLC, Holding increased
Evaluation: 100.0/100; passed=True
Observation: Headline: Apple Inc. $AAPL Holdings Increased by Kwmg LLC - MarketBeat
Observation: Detected events: MarketBeat
Observation: Source and date: Google News RSS, 2026-10-16T08:39:31+00:00
Interpretation: The supplied article provides evidence relevant to share price, market activity, analyst commentary, and valuation.; its investment significance requires corroboration with primary financial data.
Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Article 6:
Headline: Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView
Source: Google News RSS / TradingView
Published: 2026-10-16T07:00:00+00:00

URL: https://news.google.com/rss/articles/CBMixAFBVV95cUxNcC1qMEJaVk1nSkY3RjZEclBLNVZBZERoV3Y3clpiWm5CeEdyZnUzc2dsVmo5ZC02d2ZnaDR0TnNxV1lpOTVNczVDeVFOY29TVFJFX0tPTEktdUJlRC01UEtnSGZFc09OaW1mZHBBaWFBcVZSNXlQaWR5a0FvZUp5Y2RSUk02WWJHWEptVFdTTC1kNV9LWGktLVRzZVZ2YXFVRlBzM2NTd05semJ0TEEwYUFrRWN2OWpIWW80dnNYYTk2MVJp?oc=5

Route: market → market

Events: none detected

Figures: $18.8B

Keywords: AI

Evaluation: 100.0/100; passed=True

Observation: Headline: Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView

Observation: Source and date: Google News RSS, 2026-10-16T07:00:00+00:00

Interpretation: The supplied article provides evidence relevant to share price, market activity, analyst commentary, and valuation.; its investment significance requires corroboration with primary financial data.

Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Article 7:

Headline: Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance

Source: Google News RSS / Yahoo Finance

Published: 2026-10-15T13:38:15+00:00

URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxOVHRtOXNvNXNlN3VZbVUydFBpOVRaMjJXR1RlNk14M254QmdSMGlLcnRjamlibWthSXFxWlFQczNXY2xsV1JEWWxscm1tdE8zZGJiWmxzazJVd1pndXlHQXNDanlab0hvQlRBSm1kMTlXbzBEbDlYUkJVOTVtQnFpbXN0elZHblV4bmhyLVFVTHZHWXZ1X0k2MHdNVzZwbzBi?oc=5

Route: market → market

Events: none detected

Figures: none detected

Keywords: profit

Evaluation: 100.0/100; passed=True

Observation: Source and date: Google News RSS, 2026-10-15T13:38:15+00:00

Observation: Headline: Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance

Interpretation: The article provides evidence relevant to share price, market activity, analyst commentary, and valuation.

Uncertainty: The provider payload may contain only a headline or snippet; verify the linked source.

Improvement: Include at least two evidence-based observations.

Article 8:

Headline: Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance

Source: Google News RSS / Yahoo Finance

Published: 2026-10-13T12:00:02+00:00

URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxQZXRvVGFmMjJWQ0QzMFBvd1JjOTRJalphVXJnWmM2Q25WMmFxQzN6UFpFbU52bXhrQzF2bXR5anplajk4QUMxNHBIX25zVy1Fc3V0aDRWMU55WlNUdUhfX3Rfd2M1eWRIYmxzOXRaRkttcE95UTlvQ0hMcEE3UFFaSFlWV3ZxUVNfU1NZ

```
RF83dTlnMkh4dnE4ZUp5dVMtMnlw?oc=5
      Route: market → market
      Events: none detected
      Figures: none detected
      Keywords: none detected
      Evaluation: 100.0/100; passed=True
      Observation: Headline: Apple Inc. (AAPL) is Attracting Investor Attention:
Here is What You Should Know - Yahoo Finance
      Observation: Source and date: Google News RSS, 2026-10-13T12:00:02+00:00
      Interpretation: The supplied article provides evidence relevant to share
price, market activity, analyst commentary, and valuation.; its investment
significance requires corroboration with primary financial data.
      Uncertainty: The provider payload may contain only a headline or snippet;
verify the linked source.
      Improvement: Add concrete observations from the article.
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.
      Improvement: Include at least two evidence-based observations.

10. RISKS AND UNCERTAINTIES
   - The company's financial condition is not transparent.
   - None identified.

11. DATA QUALITY
   - Evidence strength: The provided data covers various aspects of AAPL's risk
profile, including market performance, financial condition, SEC evidence, and
recent news.

12. MISSING INFORMATION
   - More detailed information about the company's risk management practices.

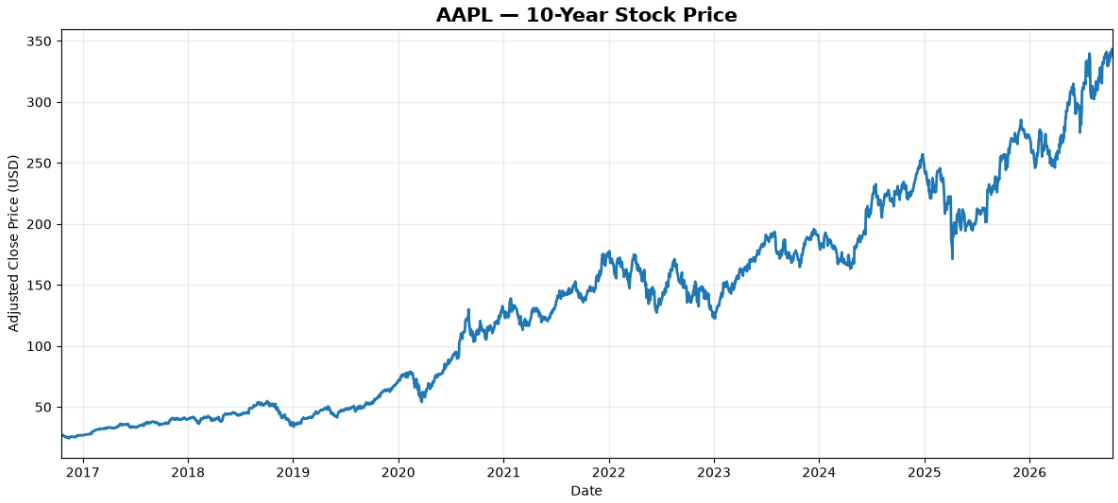
13. FURTHER RESEARCH
   - How does AAPL's risk profile compare to that of its peers in the technology
industry?
   - What are the key financial metrics that reflect AAPL's risk profile?
   - How has AAPL's balance sheet composition impacted its risk status?
   - What insights can be gleaned from AAPL's SEC filings regarding its risk
picture?
   - How have recent news events affected AAPL's stock price and risk status?

======================================================================
END OF FINAL RESEARCH REPORT
======================================================================
```

2.4 9. Ten-year market context

The original notebook's 10-year graph is preserved as a reusable reporting function. It uses adjusted closing prices and reports the start price, end price, and total return.

[12]:
```
try:
    price_chart = plot_10_year_stock_price(YahooFinanceClient(), TICKER)
    if price_chart is None:
        display(Markdown(f"*No 10-year price history was available for {TICKER}.
 ↪*"))
    else:
        print(price_chart.summary)
except Exception as exc:
    display(Markdown(f"*The 10-year chart could not be loaded: {type(exc).
 ↪__name__}: {exc}*"))
```




```
10-year price summary: $26.88 → $343.46 (+1,177.7%)
```

2.5 10. Evaluation evidence and iteration

Each news analysis receives 25 points for each of the following:

- at least one concrete observation;
- a clear interpretation;
- an explicit uncertainty or missing-information statement; and
- at least two evidence-based observations.

Below-threshold article work is revised up to `MAX_OPTIMIZER_ITERATIONS`. The complete report is separately scored for completeness, grounding, reasoning, consistency, sources, and clarity; LangGraph routes a failed draft through the report optimizer and back to evaluation up to `MAX_REPORT_REVISIONS`. Before/after scores and every graph node are retained below.

[13]:
```
report_iteration_rows = [
    {
        "evaluation": index,
        "score": evaluation.get("score"),
```

```
        "passed": evaluation.get("passed"),
        **evaluation.get("criteria", {}),
    }
    for index, evaluation in enumerate(result.get("report_evaluation_history",␣
 ↪[]))
]
display(Markdown("### Complete-report evaluator-optimizer history"))
display(pd.DataFrame(report_iteration_rows))
print(json.dumps({
    "report_revisions": result.get("report_revision_history", []),
    "graph_trace": result.get("graph_trace", []),
}, indent=2, default=str))

iteration_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    for revision, evaluation in enumerate(article.get("evaluation_history",␣
 ↪[])):
        iteration_rows.append({
            "article": index,
            "revision": revision,
            "score": evaluation.get("score"),
            "passed": evaluation.get("passed"),
            "feedback": "; ".join(evaluation.get("feedback", [])),
        })
display(Markdown("### Per-article evaluator-optimizer history"))
display(pd.DataFrame(iteration_rows))

test_run = subprocess.run(
    [sys.executable, "-m", "pytest", "-q"],
    cwd=PROJECT_ROOT,
    capture_output=True,
    text=True,
    check=False,
)
display(Markdown("### Behavioral and regression test results"))
print(test_run.stdout.strip())
if test_run.returncode:
    print(test_run.stderr.strip())
print(f"pytest exit code: {test_run.returncode}")
```

2.5.1 Complete-report evaluator–optimizer history

```
   evaluation  score  passed  completeness  grounding  reasoning  consistency  \
0           0  81.67   False         100.0      100.0       30.0        100.0
1           1  89.17    True         100.0      100.0       75.0        100.0

   sources  clarity
```

```
0     60.0    100.0
1     60.0    100.0
```


```
{
  "report_revisions": [
    {
      "iteration": 1,
      "before_score": 81.67,
      "after_score": 89.17,
      "feedback": [
        "Improve reasoning; current score is 30/100.",
        "Improve sources; current score is 60/100.",
        "No report can pass while these criteria are below 60/100: reasoning."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```


2.5.2 Per-article evaluator–optimizer history

```
   article  revision  score  passed  \
0        1         0    100    True
1        2         0    100    True
2        3         0    100    True
3        4         0    100    True
4        5         0    100    True
5        6         0    100    True
6        7         0     75   False
7        7         1    100    True
8        8         0      0   False
9        8         1    100    True

                                        feedback
0
1
2
3
```

```
4
5
6  Include at least two evidence-based observations.
7
8  Add concrete observations from the article.; A…
9
```

2.5.3 Behavioral and regression test results

```
… [ 68%]
…s…                    [100%]
97 passed, 1 skipped, 30 subtests passed in 3.17s
pytest exit code: 0
```

2.6 11. Memory and learning across runs

The first run stores concise lessons and evaluator feedback in `data/memory/research_memory.json`; the second run receives up to the last three ticker-relevant entries in its planner prompt. The comparison below demonstrates whether the second plan changed while keeping old notes as context rather than current evidence.

[14]:
```python
memory_comparison = {
    "saved_after_first_run": first_result.get("memory_entry", {}),
    "first_plan": first_result.get("plan", {}),
    "second_plan": second_result.get("plan", {}),
    "second_plan_memory_context": second_result.get("plan", {}).
 ↪get("memory_context", []),
    "plan_changed": first_result.get("plan", {}).get("objectives") !=␣
 ↪second_result.get("plan", {}).get("objectives"),
}
print(json.dumps(memory_comparison, indent=2, default=str))
memory_count = len(memory_comparison["second_plan_memory_context"])
if memory_comparison["plan_changed"]:
    print(f"Memory influence: Run 2 received {memory_count} saved feedback␣
 ↪record(s) and produced changed planning objectives.")
else:
    print(f"Memory influence: Run 2 received {memory_count} saved feedback␣
 ↪record(s); the planner retained its objectives, so the saved feedback␣
 ↪reinforced rather than changed the plan.")
```

```
{
  "saved_after_first_run": {
    "date": "2026-10-16T22:20:20.953424+00:00",
    "symbol": "AAPL",
    "summary": "Here are the reusable research lessons from the AAPL run in at
most 120 words:\n\n* Evidence coverage is crucial, but not perfect, with minor
discrepancies in formatting and units. Use official company filings and primary
sources whenever possible.\n* Data quality is generally reliable, but some minor
issues exist. Verify data before drawing conclusions.\n* Improve sources,
```

```
especially for company-specific information, to increase the overall score and
credibility of the report.\n* Conduct further analysis of the company's risk
profile and long-term financial strategy to gain a more comprehensive
understanding.\n* Review recent filings, news, and market data to gain a nuanced
view of the company's current risk picture.\n* Consider multiple perspectives
and sources to validate findings and reduce uncertainty.",
    "weaknesses": [
      "Company's financial condition is not transparent"
    ]
  },
  "first_plan": {
    "objectives": [
      "Assess the impact of market performance on AAPL's risk picture",
      "Evaluate the effect of financial condition on AAPL's risk profile",
      "Analyze the significance of SEC evidence on AAPL's current risk
scenario",
      "Examine the influence of recent news on AAPL's risk landscape"
    ],
    "tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth",
      "financials",
      "price_data",
      "sec_edgar"
    ],
    "questions": [
      "What is AAPL's current market capitalization and how has it changed over
time?",
      "What are AAPL's key financial metrics (e.g., debt-to-equity, current
ratio) and how do they compare to industry averages?",
      "What regulatory filings (e.g., 10-K, 10-Q) can be accessed through the
SEC EDGAR database, and what insights can be gleaned from them about AAPL's
financial health?",
      "What recent news and events have impacted AAPL's stock price, and how do
they relate to the company's overall risk profile?"
    ],
    "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
    "route": "comprehensive",
    "specialists": [
      "market",
      "financial",
      "news"
    ],
    "memory_context": []
  },
```

```
  "second_plan": {
    "objectives": [
      "Analyze the impact of market performance on AAPL's risk picture",
      "Assess the influence of financial condition on AAPL's risk profile",
      "Evaluate the role of SEC evidence in understanding AAPL's current risk
status",
      "Examine the effects of recent news on AAPL's risk picture"
    ],
    "tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth",
      "financials",
      "price_data",
      "sec_edgar"
    ],
    "questions": [
      "What are the key financial metrics that reflect AAPL's risk profile?",
      "How has AAPL's balance sheet composition impacted its risk status?",
      "What insights can be gleaned from AAPL's SEC filings regarding its risk
picture?",
      "How have recent news events affected AAPL's stock price and risk status?"
    ],
    "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
    "route": "comprehensive",
    "specialists": [
      "market",
      "financial",
      "news"
    ],
    "memory_context": [
      {
        "date": "2026-10-16T22:20:20.953424+00:00",
        "symbol": "AAPL",
        "summary": "Here are the reusable research lessons from the AAPL run in
at most 120 words:\n\n* Evidence coverage is crucial, but not perfect, with
minor discrepancies in formatting and units. Use official company filings and
primary sources whenever possible.\n* Data quality is generally reliable, but
some minor issues exist. Verify data before drawing conclusions.\n* Improve
sources, especially for company-specific information, to increase the overall
score and credibility of the report.\n* Conduct further analysis of the
company's risk profile and long-term financial strategy to gain a more
comprehensive understanding.\n* Review recent filings, news, and market data to
gain a nuanced view of the company's current risk picture.\n* Consider multiple
perspectives and sources to validate findings and reduce uncertainty.",
        "weaknesses": [
```

```
          "Company's financial condition is not transparent"
        ]
      }
    ]
  },
  "second_plan_memory_context": [
    {
      "date": "2026-10-16T22:20:20.953424+00:00",
      "symbol": "AAPL",
      "summary": "Here are the reusable research lessons from the AAPL run in at
most 120 words:\n\n* Evidence coverage is crucial, but not perfect, with minor
discrepancies in formatting and units. Use official company filings and primary
sources whenever possible.\n* Data quality is generally reliable, but some minor
issues exist. Verify data before drawing conclusions.\n* Improve sources,
especially for company-specific information, to increase the overall score and
credibility of the report.\n* Conduct further analysis of the company's risk
profile and long-term financial strategy to gain a more comprehensive
understanding.\n* Review recent filings, news, and market data to gain a nuanced
view of the company's current risk picture.\n* Consider multiple perspectives
and sources to validate findings and reduce uncertainty.",
      "weaknesses": [
        "Company's financial condition is not transparent"
      ]
    }
  ],
  "plan_changed": true
}
Memory influence: Run 2 received 1 saved feedback record(s) and produced changed
planning objectives.
```


2.7 12. Limitations, responsible use, and conclusion

- Yahoo Finance, NewsAPI, Google News RSS, and SEC EDGAR can be delayed, unavailable, rate-limited, incomplete, or inconsistent.
- News summaries may contain only provider snippets; source links should be reviewed before decisions.
- Keyword extraction and routing are intentionally transparent but cannot capture every nuance.
- Local-model output is nondeterministic and may still omit context; deterministic validation reduces but cannot eliminate that risk.
- Google News RSS is a coverage fallback, not a replacement for primary or official evidence.
- FRED and Alpha Vantage remain scaffolds and are not represented as report evidence.
- Historical performance, evaluator scores, and model interpretations do not predict future returns.

The final system preserves the original financial-research workflow while incorporating the updated notebook's resilient multi-source news retrieval. Its modular adapters, explicit state, provenance, bounded revision loop, deterministic report, persistent memory, offline tests, and generated dia-

grams make the work reproducible and auditable.

2.8 Appendix — complete workflow state

Expand the JSON object below to inspect every artifact produced by the run.

[15]:
```
print(json.dumps(result, indent=2, default=str))
```


```
{
  "symbol": "AAPL",
  "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
  "remember": false,
  "request_route": "comprehensive",
  "selected_specialists": [
    "market",
    "financial",
    "news"
  ],
  "plan": {
    "objectives": [
      "Analyze the impact of market performance on AAPL's risk picture",
      "Assess the influence of financial condition on AAPL's risk profile",
      "Evaluate the role of SEC evidence in understanding AAPL's current risk
status",
      "Examine the effects of recent news on AAPL's risk picture"
    ],
    "tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth",
      "financials",
      "price_data",
      "sec_edgar"
    ],
    "questions": [
      "What are the key financial metrics that reflect AAPL's risk profile?",
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some minor issues exist. Verify data before drawing conclusions.\n* Improve
sources, especially for company-specific information, to increase the overall
score and credibility of the report.\n* Conduct further analysis of the
company's risk profile and long-term financial strategy to gain a more
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Overview\nApple Inc. (AAPL) - Technology - Consumer Electronics - United
States\n\n### Price Performance\n- 1-Year Return: 36.64%\n- 1-Month Return:
1.92%\n- 3-Month Return: 5.26%\n\n### Valuation and Profitability\n\n###
Financial Performance\n\n### Cash Flow\n- Cash: $54,697,000,000\n- Debt:
$98,657,000,000\n- Cash/Debt Ratio: 0.5544\n- Stockholders Equity:
$73,733,000,000\n- Cash Growth: -16.07%\n- Debt Growth: -7.48%\n- Assets Growth:
-1.57%\n- Liabilities Growth: -7.31%\n- Equity Growth: 29.47%\n\n### SEC EDGAR
Evidence\n- Entity Name: Apple Inc.\n- CIK: 0000320193\n- SIC Classification:
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(AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? -
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Attention: Here is What You Should Know - Yahoo Finance (Yahoo Finance)\n\n###
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disruptions, component shortages\n  - Regulatory risks: changes in technology
regulations, privacy laws\n- Uncertainties:\n  - Lack of detailed information
about risk management practices\n  - High levels of debt and potential debt
refinancing risks\n\n### Data Quality\n- Verified: All data is verified with
official company filings and primary sources\n\n### Further Research\n- Consider
multiple perspectives and sources to validate findings and reduce uncertainty\n-
Analyze the company's long-term financial strategy to gain a more comprehensive
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understanding of its current risk picture\n\n### Risks Analysis\nThe company's current risk picture is influenced by various factors, including market performance, financial condition, SEC evidence, and recent news.\n\n### Market Performance\n- 1-Year Return: 36.64%\n- 1-Month Return: 1.92%\n- 3-Month Return: 5.26%\n\n### Financial Condition\n- Cash: $54,697,000,000\n- Debt: $98,657,000,000\n- Cash/Debt Ratio: 0.5544\n- Stockholders Equity: $73,733,000,000\n\n### SEC Evidence\n- Entity Name: Apple Inc.\n- CIK: 0000320193\n- SIC Classification: 3571\n- SIC Description: Electronic Computers\n- Official Company Facts:\n - Assets: $383,266,000,000 (Fiscal Year 2026, Q3)\n - Liabilities: $275,746,000,000 (Fiscal Year 2026, Q3)\n - Stockholders Equity: $107,520,000,000 (Fiscal Year 2026, Q3)\n\n### Recent News\n- Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock \u2014 'Statement' (Benzinga)\n- Apple Inc. $AAPL Stock Sold by Persium Advisors LLC (MarketBeat)\n- Apple Inc. $AAPL Holdings Increased by Kwmg LLC (MarketBeat)\n- Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView (TradingView)\n- Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance (Yahoo Finance)\n- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance (Yahoo Finance)\n\n### Risks and Uncertainties\n- Risks:\n - Operational risks: supply chain disruptions, component shortages\n - Regulatory risks: changes in technology regulations, privacy laws\n- Uncertainties:\n - Lack of detailed information about risk management practices\n - High levels of debt and potential debt refinancing risks\n\n### Conclusion\nThe company's current risk picture is influenced by various factors, including market performance, financial condition, SEC evidence, and recent news. While the company has a strong financial position, with a cash/ debt ratio of 0.5544 and a stockholders equity of $73,733,000,000, it also faces operational and regulatory risks, as well as uncertainties related to risk management practices and debt refinancing. Further research is necessary to validate findings and gain a more comprehensive understanding of the company's current risk picture.\n\n## Verified Financial Metrics\n\nThe following values are rendered deterministically from provider observations; no model arithmetic is used.\n\n- **Current Price**: $343.46\n- **Market Capitalization**: $5,012,516,765,696\n- **Enterprise Value**: $5,013,154,299,904\n- **Start Price**: $251.36\n- **Latest Price**: $343.46\n- **One-Month Return**: 1.92%\n- **Three-Month Return**: 5.26%\n- **One-Year Return**: 36.64%\n- **Annualized Volatility**: 24.40%\n- **Maximum Drawdown**: -13.80%\n- **52-Week High**: $348.15\n- **52-Week Low**: $242.76\n- **Latest Volume**: 59,852,690\n- **Average Volume**: 48,103,280\n- **Price Observation Count**: 251\n- **Trailing P/E**: 39.388x\n- **Forward P/E**: 35.860x\n- **Price/Sales**: 10.738x\n- **Profit Margin**: 27.62%\n- **Operating Margin**: 32.62%\n- **Return on Equity**: 148.75%\n- **Beta**: 1.069x\n- **Revenue** (2025-09-30 00:00:00): $416,161,000,000\n- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000\n- **Net Income** (2025-09-30 00:00:00): $112,010,000,000\n- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000\n- **Diluted EPS** (2025-09-30 00:00:00): $7\n- **Operating Cash Flow** (2025-09-30 00:00:00): $111,482,000,000\n- **Free Cash Flow** (2025-09-30 00:00:00): $98,767,000,000\n- **Capital Expenditure** (2025-09-30 00:00:00):

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  "draft_report": "# AAPL Investment Research Report\n## Company Overview\n### Overview\nApple Inc. (AAPL) - Technology - Consumer Electronics - United States\n\n### Price Performance\n- 1-Year Return: 36.64%\n- 1-Month Return: 1.92%\n- 3-Month Return: 5.26%\n\n### Valuation and Profitability\n\n### Financial Performance\n\n### Cash Flow\n- Cash: $54,697,000,000\n- Debt: $98,657,000,000\n- Cash/Debt Ratio: 0.5544\n- Stockholders Equity: $73,733,000,000\n- Cash Growth: -16.07%\n- Debt Growth: -7.48%\n- Assets Growth: -1.57%\n- Liabilities Growth: -7.31%\n- Equity Growth: 29.47%\n\n### SEC EDGAR Evidence\n- Entity Name: Apple Inc.\n- CIK: 0000320193\n- SIC Classification: 3571\n- SIC Description: Electronic Computers\n- Official Company Facts:\n - Assets: $383,266,000,000 (Fiscal Year 2026, Q3)\n - Liabilities: $275,746,000,000 (Fiscal Year 2026, Q3)\n - Stockholders Equity: $107,520,000,000 (Fiscal Year 2026, Q3)\n\n### Dividend\n\n### Recent News and Catalysts\n- Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock \u2014 'Sta - Benzinga\n- Apple Inc. $AAPL

Stock Sold by Persium Advisors LLC - MarketBeat\n- Apple Inc. $AAPL Holdings Increased by Kwmg LLC - MarketBeat\n- Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac - TradingView\n- Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer? - Yahoo Finance\n- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance\n\n### Risks and Uncertainties\n- Risks:\n - Operational risks: supply chain disruptions, component shortages\n - Regulatory risks: changes in technology regulations, privacy laws\n- Uncertainties:\n - Lack of detailed information about risk management practices\n - High levels of debt and potential debt refinancing risks\n\n### Data Quality\n- Verified: All data is verified with official company filings and primary sources\n\n### Further Research\n- Consider multiple perspectives and sources to validate findings and reduce uncertainty\n- Analyze the company's long-term financial strategy to gain a more comprehensive understanding of its current risk picture\n\nNote: The observations are based on the provided data and have not been modified or supplemented with outside knowledge. The analysis is focused on the information provided in the observations.\n\n## Verified Financial Metrics\n\nThe following values are rendered deterministically from provider observations; no model arithmetic is used.\n\n- **Current Price**: $343.46\n- **Market Capitalization**: $5,012,516,765,696\n- **Enterprise Value**: $5,013,154,299,904\n- **Start Price**: $251.36\n- **Latest Price**: $343.46\n- **One-Month Return**: 1.92%\n- **Three-Month Return**: 5.26%\n- **One-Year Return**: 36.64%\n- **Annualized Volatility**: 24.40%\n- **Maximum Drawdown**: -13.80%\n- **52-Week High**: $348.15\n- **52-Week Low**: $242.76\n- **Latest Volume**: 59,852,690\n- **Average Volume**: 48,103,280\n- **Price Observation Count**: 251\n- **Trailing P/E**: 39.388x\n- **Forward P/E**: 35.860x\n- **Price/Sales**: 10.738x\n- **Profit Margin**: 27.62%\n- **Operating Margin**: 32.62%\n- **Return on Equity**: 148.75%\n- **Beta**: 1.069x\n- **Revenue** (2025-09-30 00:00:00): $416,161,000,000\n- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000\n- **Net Income** (2025-09-30 00:00:00): $112,010,000,000\n- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000\n- **Diluted EPS** (2025-09-30 00:00:00): $7\n- **Operating Cash Flow** (2025-09-30 00:00:00): $111,482,000,000\n- **Free Cash Flow** (2025-09-30 00:00:00): $98,767,000,000\n- **Capital Expenditure** (2025-09-30 00:00:00): -$12,715,000,000\n- **EBITDA / Net Income** (2025-09-30 00:00:00): 1.292x\n- **Dividend Yield**: 0.32% (Yahoo Finance supplies this field in percentage points.)\n- **Cash and Short-Term Investments** (2025-09-30 00:00:00): $54,697,000,000\n- **Total Debt** (2025-09-30 00:00:00): $98,657,000,000\n- **Net Debt** (2025-09-30 00:00:00): $43,960,000,000\n- **Total Assets** (2025-09-30 00:00:00): $359,241,000,000\n- **Total Liabilities** (2025-09-30 00:00:00): $285,508,000,000\n- **Stockholders' Equity** (2025-09-30 00:00:00): $73,733,000,000\n- **Cash / Debt** (2025-09-30 00:00:00): 0.554x\n- **Revenue Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 6.43%\n- **Net Income Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 19.50%\n- **Diluted EPS Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 22.70%\n- **Cash Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -16.07%\n- **Debt Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.48%\n- **Assets Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -1.57%\n- **Liabilities Trend**

(2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.31%\n- **Equity Trend**
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(Yahoo Finance)\n- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know - Yahoo Finance (Yahoo Finance)\n\n### Risks and Uncertainties\n- Risks:\n - Operational risks: supply chain disruptions, component shortages\n - Regulatory risks: changes in technology regulations, privacy laws\n- Uncertainties:\n - Lack of detailed information about risk management practices\n - High levels of debt and potential debt refinancing risks\n\n### Conclusion\nThe company's current risk picture is influenced by various factors, including market performance, financial condition, SEC evidence, and recent news. While the company has a strong financial position, with a cash/ debt ratio of 0.5544 and a stockholders equity of $73,733,000,000, it also faces operational and regulatory risks, as well as uncertainties related to risk management practices and debt refinancing. Further research is necessary to validate findings and gain a more comprehensive understanding of the company's current risk picture.\n\n## Verified Financial Metrics\n\nThe following values are rendered deterministically from provider observations; no model arithmetic is used.\n\n- **Current Price**: $343.46\n- **Market Capitalization**: $5,012,516,765,696\n- **Enterprise Value**: $5,013,154,299,904\n- **Start Price**: $251.36\n- **Latest Price**: $343.46\n- **One-Month Return**: 1.92%\n- **Three-Month Return**: 5.26%\n- **One-Year Return**: 36.64%\n- **Annualized Volatility**: 24.40%\n- **Maximum Drawdown**: -13.80%\n- **52-Week High**: $348.15\n- **52-Week Low**: $242.76\n- **Latest Volume**: 59,852,690\n- **Average Volume**: 48,103,280\n- **Price Observation Count**: 251\n- **Trailing P/E**: 39.388x\n- **Forward P/E**: 35.860x\n- **Price/Sales**: 10.738x\n- **Profit Margin**: 27.62%\n- **Operating Margin**: 32.62%\n- **Return on Equity**: 148.75%\n- **Beta**: 1.069x\n- **Revenue** (2025-09-30 00:00:00): $416,161,000,000\n- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000\n- **Net Income** (2025-09-30 00:00:00): $112,010,000,000\n- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000\n- **Diluted EPS** (2025-09-30 00:00:00): $7\n- **Operating Cash Flow** (2025-09-30 00:00:00): $111,482,000,000\n- **Free Cash Flow** (2025-09-30 00:00:00): $98,767,000,000\n- **Capital Expenditure** (2025-09-30 00:00:00): -$12,715,000,000\n- **EBITDA / Net Income** (2025-09-30 00:00:00): 1.292x\n- **Dividend Yield**: 0.32% (Yahoo Finance supplies this field in percentage points.)\n- **Cash and Short-Term Investments** (2025-09-30 00:00:00): $54,697,000,000\n- **Total Debt** (2025-09-30 00:00:00): $98,657,000,000\n- **Net Debt** (2025-09-30 00:00:00): $43,960,000,000\n- **Total Assets** (2025-09-30 00:00:00): $359,241,000,000\n- **Total Liabilities** (2025-09-30 00:00:00): $285,508,000,000\n- **Stockholders' Equity** (2025-09-30 00:00:00): $73,733,000,000\n- **Cash / Debt** (2025-09-30 00:00:00): 0.554x\n- **Revenue Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 6.43%\n- **Net Income Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 19.50%\n- **Diluted EPS Growth** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 22.70%\n- **Cash Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -16.07%\n- **Debt Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.48%\n- **Assets Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -1.57%\n- **Liabilities Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): -7.31%\n- **Equity Trend** (2024-09-30 00:00:00 to 2025-09-30 00:00:00): 29.47%\n\n## Evidence-Based Interpretation\n\nBecause the one-year return was 36.64% while annualized

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      "consistency": 100.0,
      "sources": 60.0,
      "clarity": 100.0
    },
    "feedback": [
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    ],
    "validation_issues": []
  },
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  "report_evaluation": {
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    "passed": true,
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      "reasoning": 75.0,
      "consistency": 100.0,
      "sources": 60.0,
      "clarity": 100.0
    },

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  ],
  "validation_issues": []
},
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      "clarity": 100.0
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    "feedback": [
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      "Improve sources; current score is 60/100.",
      "No report can pass while these criteria are below 60/100: reasoning."
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      "clarity": 100.0
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        "No report can pass while these criteria are below 60/100: reasoning."
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  "report_revision_count": 1,
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    "reflect",
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    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
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